Memory

would like to have a computer memory that is both very large and very fast ... but there's a *fundamental tradeoff* between size and speed

solution is to use multiple components of varying sizes and speeds, in a way that (ideally!) achieves the benefits of *both* high speed *and* large size

for this purpose, implement the "memory space" abstraction in machine/assembly language using a *memory hierarchy*, consisting of:

- processor cache memory (fastest but smallest, implemented in SRAM)
- main memory (DRAM)
- secondary memory (magnetic disk, SSD, ... large but relatively very slow)

SRAM implemented using flip-flop type circuits, typically 6 transistors per bit; data retained as long as power is on

DRAM much less area per bit (one capacitor, one transistor) but slower access time; use periodic "refresh" to retain data